

VAPO

Value-Model-Based RL for Advanced Reasoning

60.4

AIME 2024 avg@32

State-of-the-Art

Base model: **Qwen2.5-32B (no SFT)** · Team: **ByteDance Seed**

+10+ pts vs DAPO · No SFT data · 5,000 training steps

arxiv.org/abs/2504.05118

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-FREE

GRPO · DAPO

- Trajectory-level reward assigned uniformly to all tokens
- No distinction between good and bad intermediate steps
- Credit assignment is coarse — token-level errors go undetected
- Simpler to train, but limited for complex multi-step reasoning

VS

Value-Model-BASED

VAPO (PPO family)

- Precise token-level credit assignment via learned value function
- Traces each action's impact on final return
- Critical when subtle step-level errors cause catastrophic failures
- Lower-variance estimates per token vs Monte Carlo methods

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

01 Value Model Bias

Long trajectories + bootstrapped learning

In long-CoT tasks, value estimates must propagate through hundreds of tokens. Bootstrapped TD learning compounds errors, introducing systematic bias that distorts credit assignment.

02 Heterogeneous Sequence Lengths

Different bias-variance tradeoffs needed

Responses vary dramatically in length. A single fixed λ in GAE cannot simultaneously minimize variance for short outputs and bias for long outputs — the optimizer must trade one off against the other.

03 Reward Signal Sparsity

Only the final <eos> token receives reward

Binary final-answer rewards are assigned at a single terminal step. Over a 6,000-token trajectory, this sparse signal must propagate backwards through every reasoning step, severely degrading the learning signal.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Challenge 1 Value Model Bias

Challenge 2 Heterogeneous Lengths

Challenge 3 Reward Sparsity

Value-Pretraining

Novel (VAPO)

Pre-trains value model before RL to reduce initial bias

Decoupled-GAE

from VC-PPO

Separates GAE computation for robust value estimation

Length-Adaptive GAE

Novel (VAPO)

Dynamic λ per response length — low λ for short, high λ for long

Clip-Higher

from DAPO

Asymmetric clipping to encourage exploration

Token-level Loss

from DAPO

Normalizes loss at token level rather than sequence level

Positive Example LM Loss

from GRPO

Self-imitation learning on correct trajectories

Group-Sampling

from GRPO

Samples multiple responses per prompt for better baseline

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

SHORT Responses

$\lambda = \text{LOW}$

(e.g. $\lambda \rightarrow 0$)

Lower Variance

Priority: Variance Reduction

Short trajectories have fewer compounding errors. A lower λ discounts future rewards faster, relying more on the immediate TD error — reducing variance at the cost of some bias.

LONG Responses

$\lambda = \text{HIGH}$

(e.g. $\lambda \rightarrow 1$)

Lower Bias

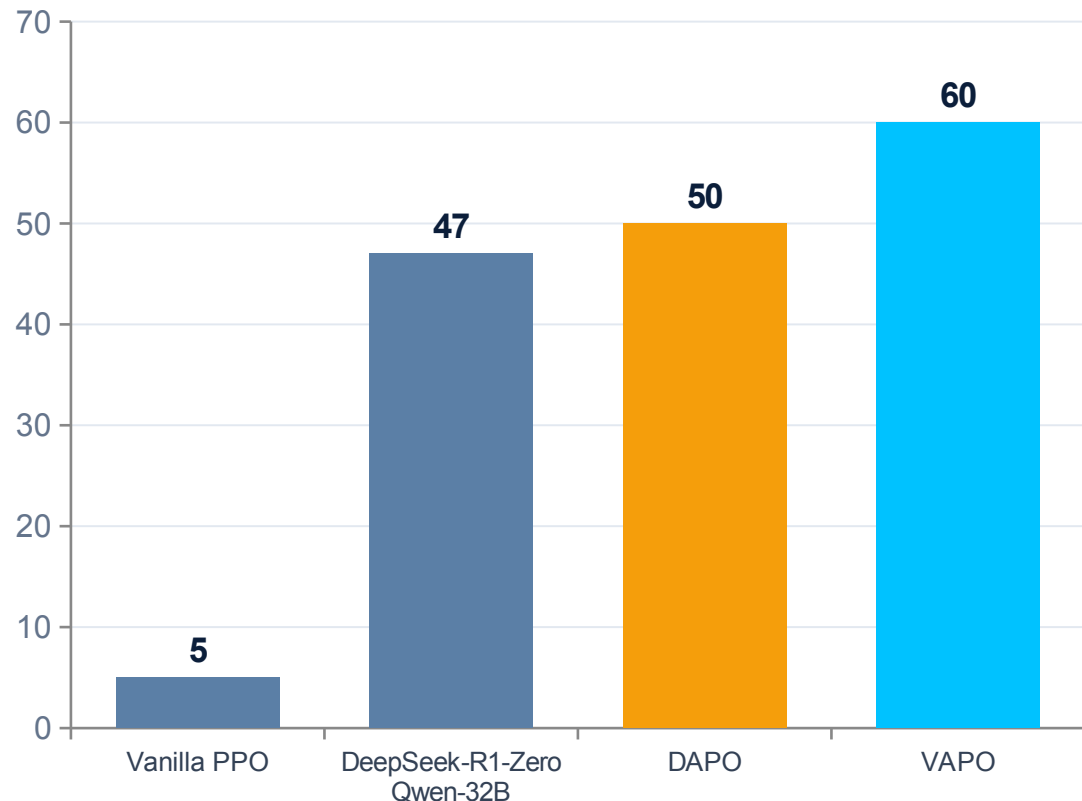
Priority: Bias Reduction

Long trajectories suffer from value model bias due to bootstrapping over many steps. Higher λ uses more Monte Carlo returns, reducing bias at the cost of higher variance.

Bias-Variance
Tradeoff

Key Insight: Length-adaptive GAE is the only technique specifically designed for heterogeneous-length CoT responses, contributing +15 points to VAPO's final score.

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



KEY RESULTS

60.4

VAPO AIME 2024 (avg@32)

+10 pts

above DAPO (50)

+13 pts

above DeepSeek-R1-Zero (47)

~5,000

steps to reach SOTA

Fig. 2: Training curves → Slide 7

VAPO Reaches SOTA in ~5K Steps — Stable Training, Longer Reasoning Chains

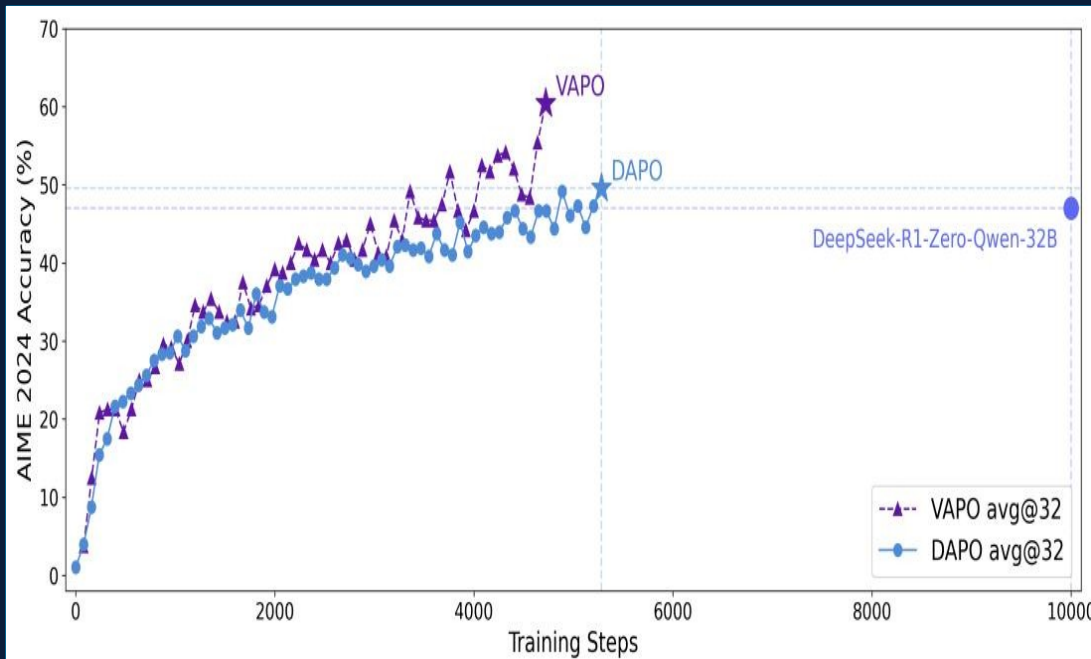


Figure 2 · AIME 2024 Accuracy vs Training Steps

VAPO hits 60% accuracy at step ~4,500 vs DAPO's ~5,500 for only 50%

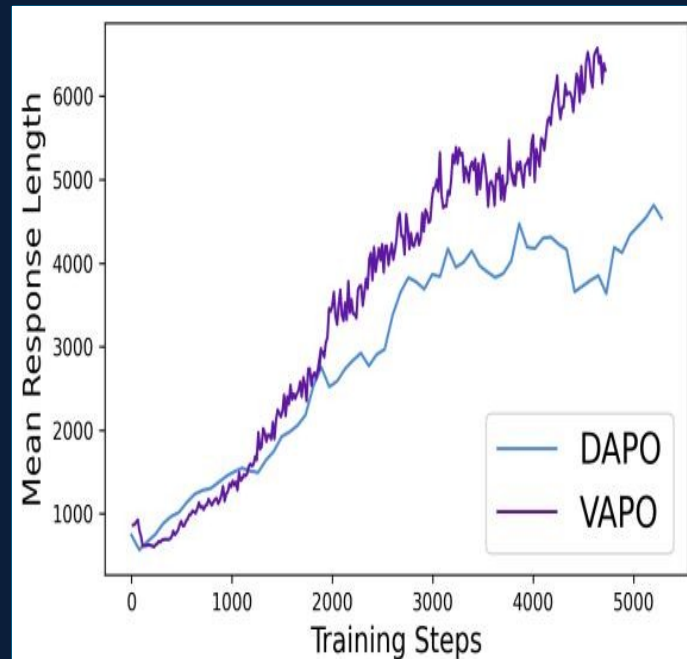
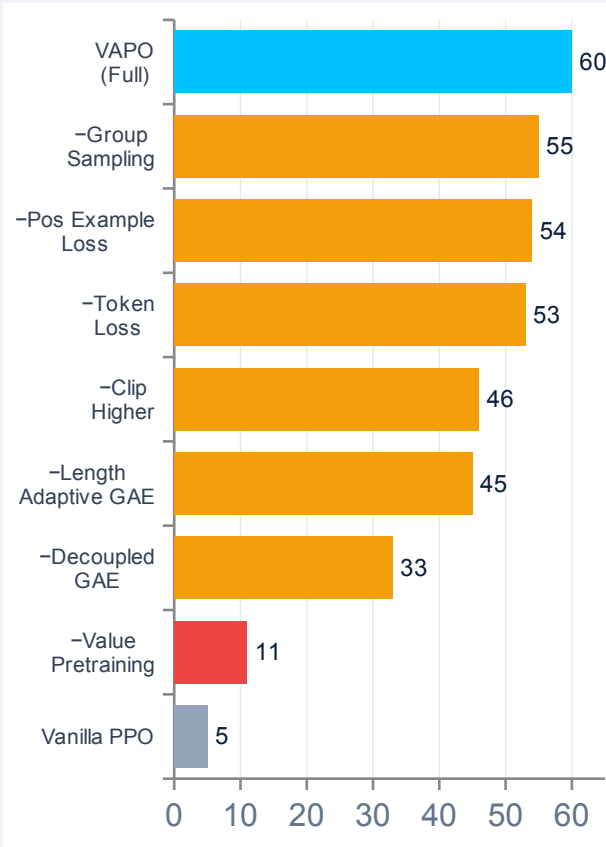


Figure 3 · Mean Response Length (tokens) vs Steps

Zero training crashes across multiple independent runs — DAPO is less stable

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Model / Ablation	AIME24	Drop
Vanilla PPO	5	—
VAPO w/o Value-Pretraining	11	-49
VAPO w/o Decoupled-GAE	33	-27
VAPO w/o Length-adaptive GAE	45	-15
VAPO w/o Clip-Higher	46	-14
VAPO w/o Token-level Loss	53	-7
VAPO w/o Positive Example Loss	54	-6
VAPO w/o Group-Sampling	55	-5
VAPO (Full)	60	—
Removing Value-Pretraining is catastrophic: score crashes from 60 to 11 (-49 pts). Value pretraining is the single most critical component.		



Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1 Value-model RL can surpass value-free methods

VAPO (60.4) beats DAPO (50) and DeepSeek-R1-Zero (47) — all on the same Qwen2.5-32B base, no SFT — proving value-model-based RL can outperform policy-gradient methods when engineered properly.

2 Value-Pretraining: the single most critical component

Removing it collapses score from 60 to 11 (−49 pts). Cold-starting the value model is nearly equivalent to Vanilla PPO. Value models must be initialized well before RL begins.

3 Length-Adaptive GAE: novel +15 point contribution

Dynamically tunes the bias-variance tradeoff per response length. Addresses a problem unique to long-CoT settings that prior work ignored — short responses use low λ , long responses use high λ .

4 Training-efficient and crash-free

Reaches SOTA in ~5,000 steps with zero crashes across multiple independent runs, outpacing DAPO which needs ~5,500 steps for a lower score.

5 A roadmap for future reasoning model development

The three core challenges (bias, heterogeneous lengths, reward sparsity) are solvable. VAPO's ablation provides a clear engineering roadmap for pushing reasoning frontiers further.